

LAB # 8**USING FUNCTIONS IN C++****TASKS TO BE PERFORMED**

1. Using function, write a complete program that prints your name 10 times. The Function can take no arguments and should not return any value.
2. Write a function to calculate the factorial value of any integer entered through the keyboard.
3. Write a function which receives a float and an int from main (), finds the product of these two and returns the product which is printed through main ().
4. Given three variables x, y, z write a function to circularly shift their values to right. In other words if x = 5, y = 8, z = 10 after circular shift y = 5, z = 8, x = 10 after circular shift y = 5, z = 8 and x = 10. Call the function with variables a, b, c to circularly shift values.
5. Identify the errors (if any) in your own words in the given pieces of code:

```
a)
func(int a,float b)
{
return (5.75);
}
```

```
b)
int , float newfunction(void)
{
return (7, 6.96);
}
```

6. Write a program that ask for three numbers, compare them and show the maximum. Declare a function called max_three that compares the numbers and returns the maximum.
7. Write a program that asks the user for an integer number and find the sum of all natural numbers upto that number.

1. Using function, write a complete program that prints your name 10 times. The Function can take no arguments and should not return any value.

```
#include <iostream>
using namespace std;

void name() {
    cout << "Muhammad Ali"<<endl;
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023"<<endl;
    for (int i = 1; i <= 10; i++) {
        cout << i<<" ";
        name();
    }
}
```



Microsoft Visual Studio Debug Console



```
Muhammad Ali
2025F-BIT-023
1)Muhammad Ali
2)Muhammad Ali
3)Muhammad Ali
4)Muhammad Ali
5)Muhammad Ali
6)Muhammad Ali
7)Muhammad Ali
8)Muhammad Ali
9)Muhammad Ali
10)Muhammad Ali
```

2. Write a function to calculate the factorial value of any integer entered through the keyboard.

```
#include <iostream>
using namespace std;
int factorial(){
    int fac = 1, x, i = 1;
    cout << "Enter number for factorial: ";
    cin >> x;
    while (i <= x) {
        fac *= i;
        i++;
    }
    cout << "factorial of "<<x<<" is "<<fac;
    return 0;
}
int main() {
    cout << "Muhammad Ali\n2025F-BIT-023"<<endl;
    factorial();
}
```

Microsoft Visual Studio Debug Console

Muhammad Ali
2025F-BIT-023
Enter number for factorial: 6
factorial of 6 is 720
C:\Users\DELL\source\repos\ConsoleApp1\

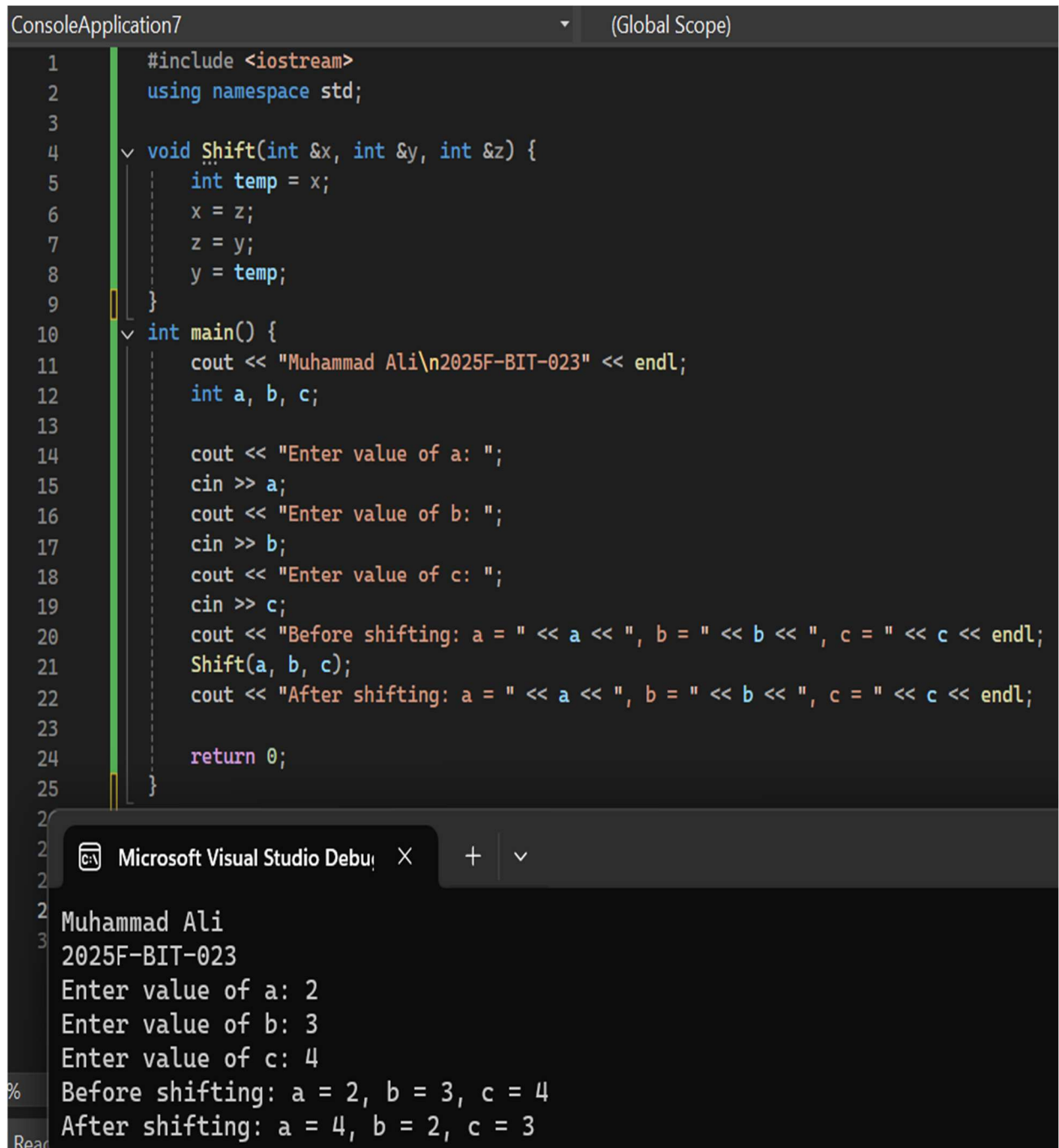
3. Write a function which receives a float and an int from main (), finds the product of these two and returns the product which is printed through main ().

```
1  #include <iostream>
2  using namespace std;
3
4  double product(float a,int b) {
5      return a * b;
6  }
7
8  int main() {
9      cout << "Muhammad Ali\n2025F-BIT-023"<<endl;
10     float a;
11     int b;
12     cout << "enter float: ";
13     cin >> a;
14     cout << "enter integer: ";
15     cin >> b;
16     cout << "product of integer and float is: " << product(a, b);
17     return 0;
18 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
enter float: 3.3
enter integer: 3
product of integer and float is: 9.9
```

4. Given three variables x, y, z write a function to circularly shift their values to right. In other words if x = 5, y = 8, z = 10 after circular shift y = 5, z = 8, x = 10 after circular shift y = 5, z = 8 and x = 10. Call the function with variables a, b, c to circularly shift values.



```
ConsoleApplication7 (Global Scope)
1  #include <iostream>
2  using namespace std;
3
4  void Shift(int &x, int &y, int &z) {
5      int temp = x;
6      x = z;
7      z = y;
8      y = temp;
9  }
10 int main() {
11     cout << "Muhammad Ali\n2025F-BIT-023" << endl;
12     int a, b, c;
13
14     cout << "Enter value of a: ";
15     cin >> a;
16     cout << "Enter value of b: ";
17     cin >> b;
18     cout << "Enter value of c: ";
19     cin >> c;
20     cout << "Before shifting: a = " << a << ", b = " << b << ", c = " << c << endl;
21     Shift(a, b, c);
22     cout << "After shifting: a = " << a << ", b = " << b << ", c = " << c << endl;
23
24     return 0;
25 }
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
Enter value of a: 2
Enter value of b: 3
Enter value of c: 4
Before shifting: a = 2, b = 3, c = 4
After shifting: a = 4, b = 2, c = 3
```

5. identify the errors (if any) in your own words in the given pieces of code:

```
a)func(int a,float b)
{
    return (5.75);
}
```

Error:

- 1) Func datatype is not define
- 2) in parameters two variables are declared and only one value is returned.
- 3) And while returning first value should be integer and second value should be float

Correct code:

```
int func(int a, float b)
{
    return (3,5.75);
}
```

```
b)
int , float newfunction(void)
{
    return (7, 6.96);
}
```

Error:

- 1) Before function name two datatype is given which should be one

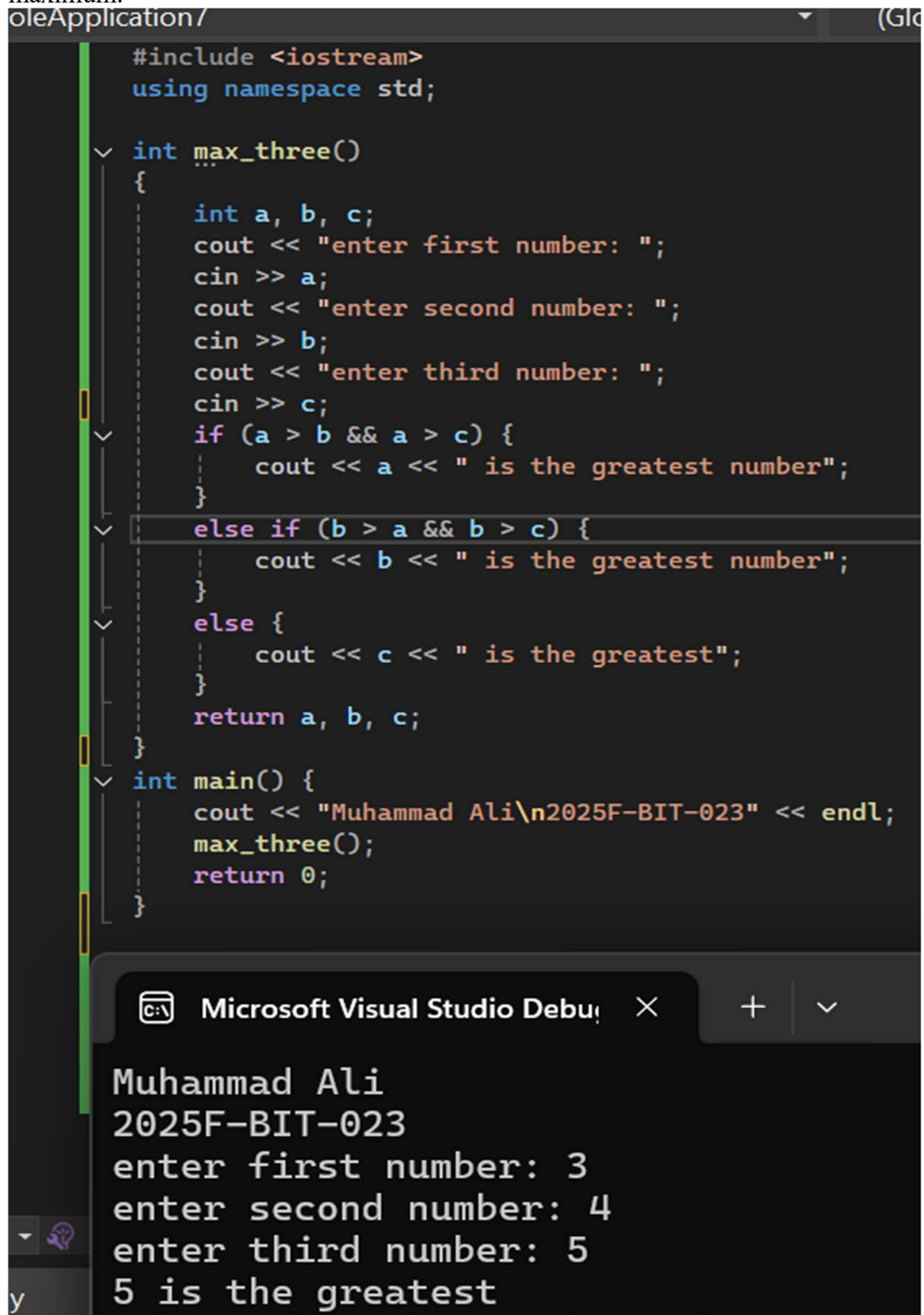
Correct code:

```
int newfunction(void)
{
    return (7, 6.96);
}
```

OR

```
float newfunction(void)
{
    return (7, 6.96);
}
```

6. Write a program that ask for three numbers, compare them and show the maximum. Declare a function called max_three that compares the numbers and returns the maximum.



```
oleApplication/ (Glo
#include <iostream>
using namespace std;

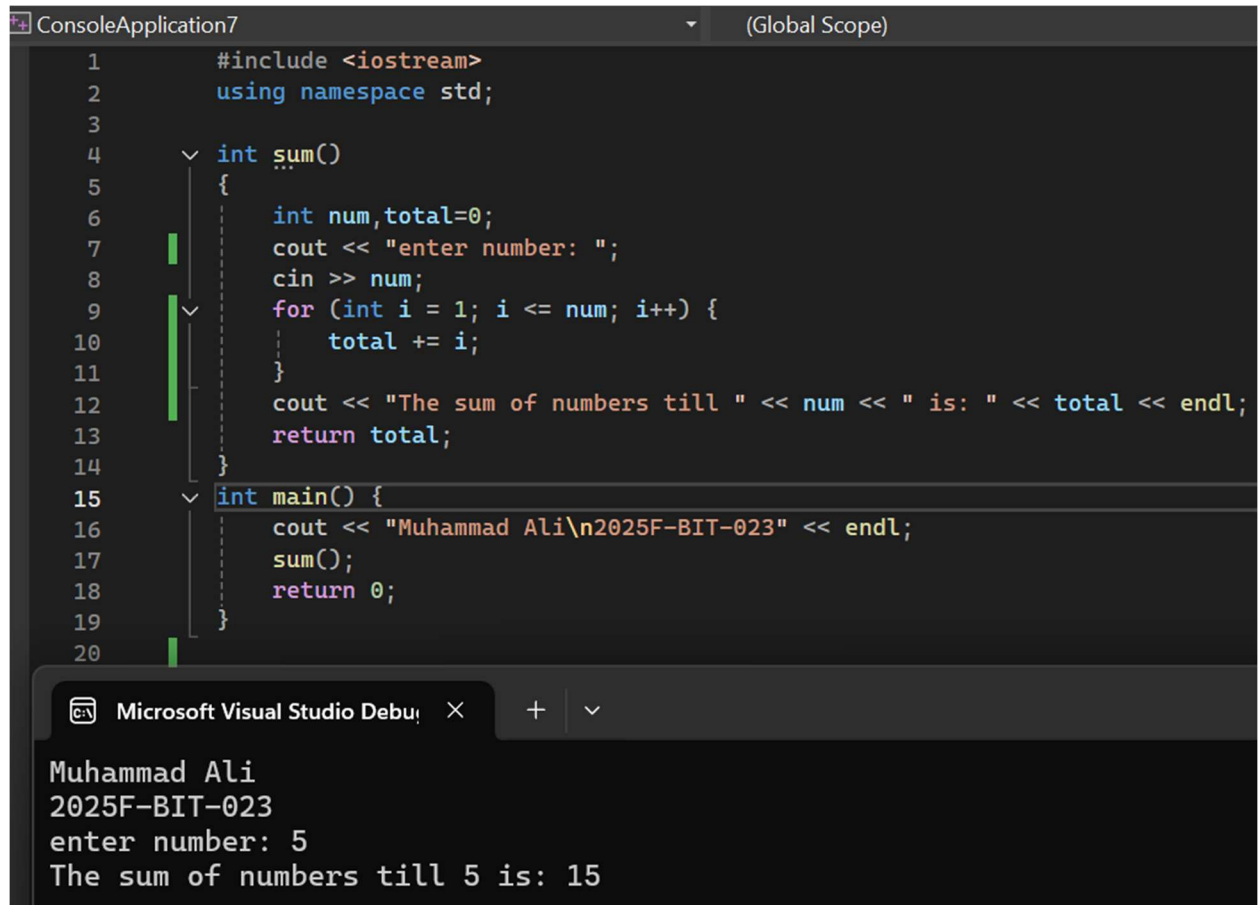
int max_three()
{
    int a, b, c;
    cout << "enter first number: ";
    cin >> a;
    cout << "enter second number: ";
    cin >> b;
    cout << "enter third number: ";
    cin >> c;
    if (a > b && a > c) {
        cout << a << " is the greatest number";
    }
    else if (b > a && b > c) {
        cout << b << " is the greatest number";
    }
    else {
        cout << c << " is the greatest";
    }
    return a, b, c;
}

int main() {
    cout << "Muhammad Ali\n2025F-BIT-023" << endl;
    max_three();
    return 0;
}
```

Microsoft Visual Studio Debug Console

```
Muhammad Ali
2025F-BIT-023
enter first number: 3
enter second number: 4
enter third number: 5
5 is the greatest
```

7. Write a program that asks the user for an integer number and find the sum of all natural numbers upto that number.



```
1  #include <iostream>
2  using namespace std;
3
4  int sum()
5  {
6      int num, total=0;
7      cout << "enter number: ";
8      cin >> num;
9      for (int i = 1; i <= num; i++) {
10         total += i;
11     }
12     cout << "The sum of numbers till " << num << " is: " << total << endl;
13     return total;
14 }
15 int main() {
16     cout << "Muhammad Ali\n2025F-BIT-023" << endl;
17     sum();
18     return 0;
19 }
20
```

Microsoft Visual Studio Debug Console Output:

```
Muhammad Ali
2025F-BIT-023
enter number: 5
The sum of numbers till 5 is: 15
```